

CASE 12

MHC CLASS I DEFICIENCY

A failure of antigen processing.

The class I molecules encoded by the major histocompatibility complex (MHC) are expressed to a greater or lesser extent on the surface of all the cells of the body except the red blood cells. MHC class I molecules bind peptides derived from proteins synthesized in the cytoplasm, and carry them to the cell surface, where they form a complex of peptide and MHC molecule on the cell surface. This complex can then be recognized by antigen-specific CD8⁺ T cells. T cells as a class recognize only peptides presented to them as a complex with an MHC molecule; the T-cell receptors of CD8⁺ T cells recognize only peptides presented by MHC class I molecules, whereas those of CD4⁺ T cells recognize only peptides presented by MHC class II molecules (see Case 8).

MHC class I molecules are involved principally in immune reactions against virus infections. Cytotoxic CD8⁺ T cells specific for viral antigens terminate viral infections by recognizing viral peptides carried by MHC class I molecules on the surface of virus-infected cells, and then killing these cells (Fig. 12.1). They release the pore-forming protein perforin and cytotoxic granzymes, as well as the pro-inflammatory cytokines tumor necrosis factor α (TNF- α) and interferon- γ (IFN- γ). In addition, cytotoxic T cells express the Fas ligand, which engages the cell-surface molecule Fas on target cells. Both processes induce the target cells to undergo programmed cell death (apoptosis).

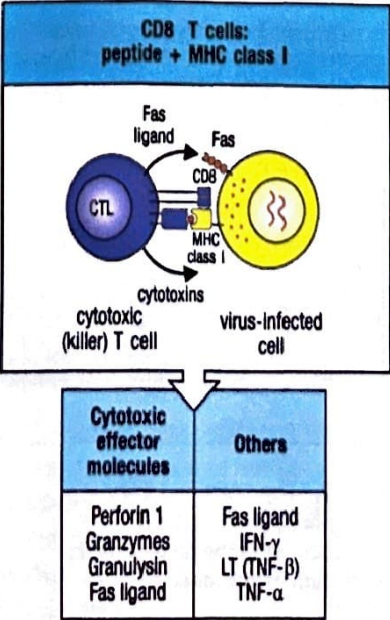


Fig 12.1 Cytotoxic CD8 T cells recognize and kill virus-infected cells. CD8⁺ T cells recognize virally derived peptides presented by MHC class I antigens at the surface of virus-infected cells. They kill target cells by releasing cytotoxins and cytotoxic cytokines and by binding to Fas on the target cell. IFN- γ , interferon- γ .

- TOPICS BEARING ON THIS CASE:**
- Genetic organization of the MHC
 - Processing of intracytoplasmic protein antigens
 - Transport of MHC class I molecules to the cell surface
 - MHC class I molecules and CD8 T-cell intrathymic maturation
 - Effector CD8 T-cell function in virus infection
 - FACS analysis

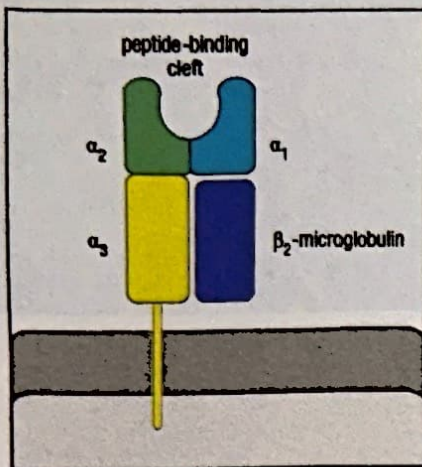


Fig. 12.2 A schematic representation of a human MHC class I molecule. It is a heterodimeric glycoprotein, composed of one transmembrane α chain bound noncovalently to β_2 -microglobulin. The α chain is folded into three protein domains, two of which form a cleft into which the peptide antigen binds.

In humans, the class I and class II MHC molecules are known as the human leukocyte antigens (HLA), and together they determine the tissue type of an individual. Each individual expresses three principal types of class I molecule—HLA-A, HLA-B, and HLA-C. These are heterodimeric glycoproteins, composed of an α chain and a β chain, the latter known also as β_2 -microglobulin (Fig. 12.2).

The genes encoding the α chains of the human MHC class I molecules are located close together in the MHC on the short arm of chromosome 6 (Fig. 12.3). In humans, the gene encoding β_2 -microglobulin, the polypeptide chain common to all class I molecules, is located not in the MHC but on the long arm of chromosome 15. The genes encoding the α chains are highly polymorphic, and so there are numerous variants of HLA-A, HLA-B, and HLA-C within the population.

Viral proteins, like cellular proteins, are made in the cytoplasm of the infected cell, and some are soon degraded into peptide fragments by large enzyme complexes called proteasomes. The peptides are then transported from the cytosol into the endoplasmic reticulum by a complex of two transporter proteins, called TAP1 and TAP2, that is located in the endoplasmic reticulum membrane (Fig. 12.4). The genes encoding TAP1 and TAP2 are also located in the MHC, in the region containing the class II genes (see Fig. 12.3).

The endoplasmic reticulum contains MHC class I molecules, which enter as separate α and β chains as soon as they have been synthesized and are retained there. After the antigenic peptides enter the endoplasmic reticulum they are loaded onto the complex of α chains and β_2 -microglobulin. The TAP-binding protein (also known as TAPBP or tapasin) facilitates the interaction of MHC class I molecules with TAP1 and TAP2, and promotes loading of antigenic peptides into this complex. The peptide:MHC class I complex is then released and transported onward to the cell surface (Fig. 12.5). In humans, the *TAPBP* gene is also located within the MHC cluster on the short arm of chromosome 6.

This case describes a rare inherited immune deficiency accompanied by the absence of MHC class I molecules on the patients' cells.

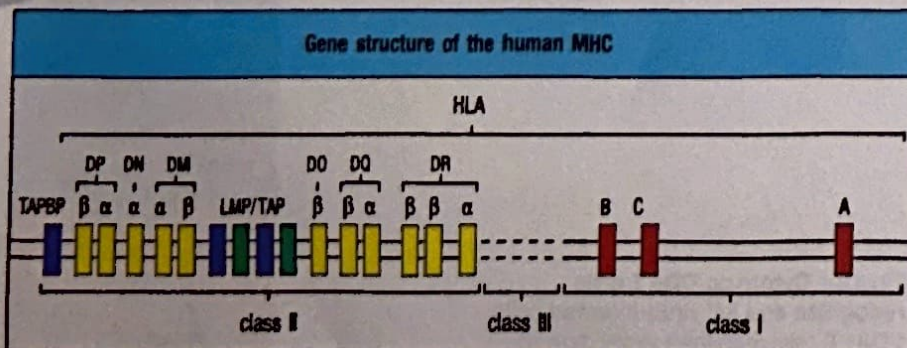


Fig. 12.3 The organization of the major histocompatibility complex (MHC) on chromosome 6 in humans. There are separate regions of class I and class II genes. The class I genes are called HLA-A, HLA-B, and HLA-C. The gene for β_2 -microglobulin is located on chromosome 15. The genes for the TAP1:TAP2 transporter are located in the class II region of the MHC.

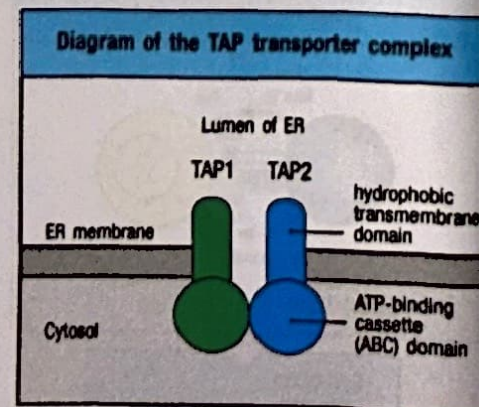


Fig. 12.4 The TAP1 and TAP2 transporter proteins form a heterodimer in the endoplasmic reticulum membrane.

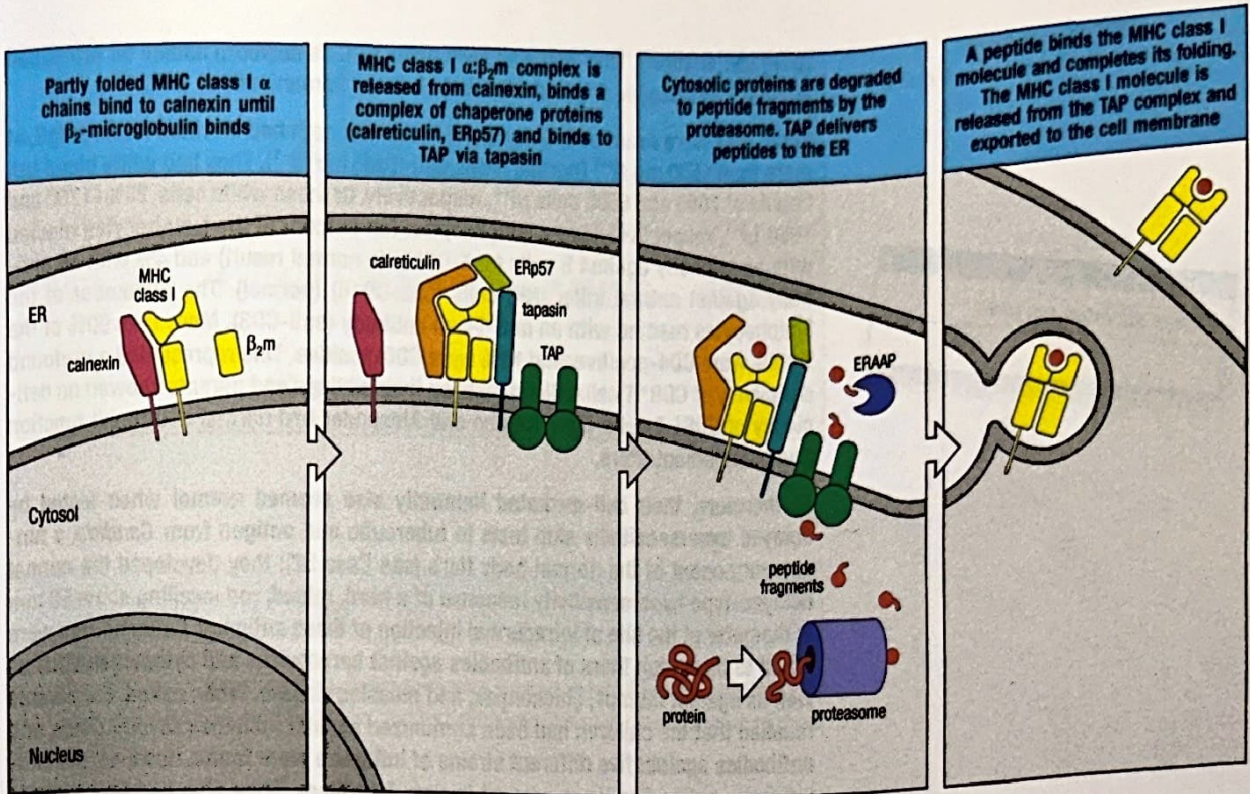


Fig. 12.5 MHC class I molecules do not leave the endoplasmic reticulum unless they bind peptides. MHC class I α chains assemble in the endoplasmic reticulum with the membrane-bound protein, calnexin. When this complex binds β_2 -microglobulin (β_2m), the MHC class I α : β_2m dimer is released from calnexin, and the partly folded MHC class I molecule then binds to the TAP1 subunit of the TAP transporter by interacting with one molecule of

the TAP-associated protein, tapasin. The MHC class I molecule is retained within the endoplasmic reticulum until it binds a peptide. Peptides generated by the degradation of proteins in the cytoplasm are transported into the lumen of the endoplasmic reticulum by the TAP transporter. Once peptide has bound to the MHC molecule, the peptide:MHC complex is transported through the Golgi complex to the cell surface.

The children of Sergei and Natasha Islayev: the consequences of a small flaw in the MHC.

Tatiana Islayev was 17 years old when first seen at the Children's Hospital. She had severe bronchiectasis (dilatation of the bronchi from repeated infections) and a persistent cough that produced yellow-green sputum. Tatiana had been chronically ill from the age of 4 years, when she started to get repeated infections of the sinuses, middle ears, and lungs, apparently due to a variety of respiratory viruses. The bacteria *Haemophilus influenzae* and *Streptococcus pneumoniae* could be cultured from her sputum, and she had been prescribed frequent antibiotic treatment to control her persistent fevers and cough. Her brother Alexander, aged 7 years, also suffered from chronic respiratory infections. Like his sister, he had begun to suffer from severe repeated viral infections of the upper and lower respiratory tracts at an early age. He also had severe bronchiectasis, and *H. influenzae* could be cultured from his sputum.

Owing to the chronic illness of Tatiana and Alexander, the Islayevs had emigrated recently from Russia to the United States, where they hoped to get better medical treatment. When they came to America they had three other children, aged 5, 10, and 13 years, who were all healthy and showed no increased susceptibility to infection. As infants in Moscow, both Tatiana and Alexander had received routine immunizations with oral poliovirus as well as diphtheria, tetanus, and pertussis

Brother and sister with
symptoms of severe chronic
respiratory infection.

Other siblings normal.

(DTP) vaccinations. They had also been given BCG as newborn babies for protection against tuberculosis, and had tolerated all these immunizations well.

When they were examined, Tatiana and Alexander both had elevated levels of IgG, at more than 1500 mg dl^{-1} (normal levels $600\text{--}1400 \text{ mg dl}^{-1}$). They had white blood cell counts of 7000 and 6600 cells μl^{-1} , respectively. Of these white cells, 25% (1750 and 1650 μl^{-1} , respectively) were lymphocytes. Ten percent of the lymphocytes reacted with an antibody against B cells (anti-CD19) (a normal result) and 4% with an antibody against natural killer (NK) cells (anti-CD16) (normal). The remainder of the lymphocytes reacted with an anti-T-cell antibody (anti-CD3). More than 90% of the T cells were CD4-positive, and 10% were CD8-positive. This represented a profound deficiency of CD8⁺ T cells. Blood tests on their siblings and parents showed no deficiency of CD8⁺ T cells. Both Tatiana and Alexander had normal neutrophil function and complement titers.

Furthermore, their cell-mediated immunity also seemed normal when tested by delayed hypersensitivity skin tests to tuberculin and antigen from *Candida*, a fungal component of the normal body flora (see Case 52); they developed the normal delayed-type hypersensitivity response of a hard, raised, red swelling some 50 mm in diameter at the site of intradermal injection of these antigens. Both children were found to have high titers of antibodies against herpesvirus and cytomegalovirus as well as against mumps, chickenpox, and measles viruses. When asked, the parents recalled that the children had been immunized against influenza several times, and antibodies against five different strains of influenza were found. However, the anti-influenza antibodies were present in very low titers. They also had low titers of antibody against Epstein-Barr virus.

When white blood cells from all family members were typed for HLA antigens by a serological typing assay, no MHC class I molecules at all could be found on Tatiana's and Alexander's cells. When their blood cells were examined using the more sensitive technique of FACS analysis (see Fig. 1.3), it was found that Tatiana and Alexander expressed very small amounts of MHC class I molecules, less than 1% of the amount expressed on the cells of their father (Fig. 12.6). In contrast, they expressed MHC class II molecules normally.

The HLA typing revealed that the mother and father shared an MHC haplotype (HLA-A3, -B63, HLA-DR4, -DQ3) and that Tatiana and Alexander had inherited this shared haplotype from both their parents (Fig. 12.7). They were therefore homozygous for the MHC region. The other children were heterozygous. It was thus concluded that the two children's susceptibility to respiratory infections was linked to the MHC locus, for which only they were homozygous.

To try to determine the underlying defect, B cells from Tatiana and Alexander were established as cell lines in culture by transformation with Epstein-Barr virus. The transformed B cells were examined for the presence of messenger RNA for MHC class I molecules; normal levels were found. This eliminated the possibility that they had a structural or regulatory defect in genes encoding the α chain of the MHC class I molecules. Because the gene for β_2 -microglobulin maps to chromosome 15, it was highly unlikely that their MHC-linked condition resulted from a defect in that gene. Only one possibility remained to be explored—a mutation in the *TAP1*, *TAP2*, or *TAPBP* genes. When the DNA sequences of these genes were determined, both Tatiana and Alexander were found to have the same nonsense mutation in their *TAP2* genes. Their parents were found to be heterozygous for this mutation.

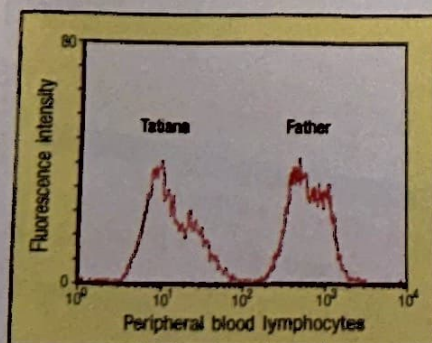


Fig. 12.6 Fluorescent antibody typing for HLA antigens. The peripheral blood lymphocytes of Tatiana and her father were typed by a fluorescent antibody against the MHC class I molecule HLA-A3. Tatiana's lymphocytes express about 1% of the HLA-A3 expressed on the lymphocytes of her father.

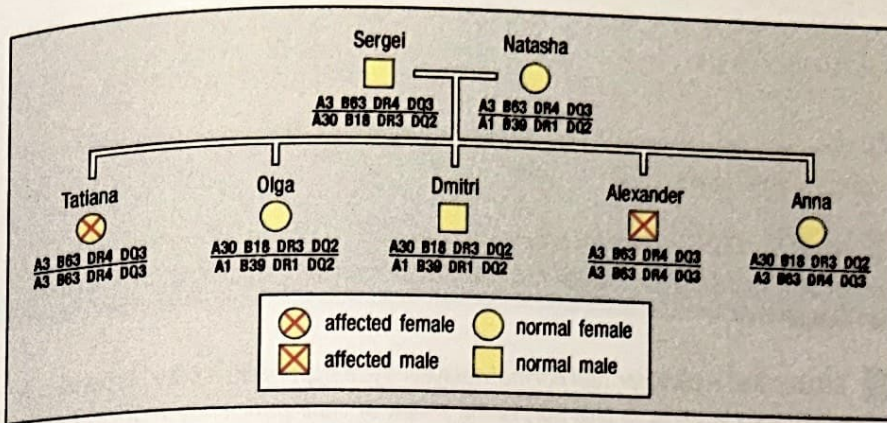


Fig. 12.7 Inheritance of MHC haplotypes in the Islayev family.

MHC class I deficiency.

A recessively inherited immunodeficiency was first suggested by the similarity of Tatiana's and Alexander's condition and its complete absence from other members of the family.

The clinical phenotype observed in the Islayev children and associated with MHC class I deficiency has some unexpected features. Despite a profound deficiency in the number of CD8⁺ T cells and the inability to present viral antigens to CD8⁺ T cells because of the absence of MHC class I antigens, the Islayev children were apparently able to fight some viral infections. The high levels of antibodies against chickenpox, measles, and mumps viruses in their blood showed that they had been exposed to and successfully overcome these infections.

They had sustained innumerable respiratory viral infections, however, and their poor antibody responses to a variety of influenza strains showed that they might have had problems responding to respiratory viruses in general. It might have been that some viruses were better able than others to stimulate an increased expression of MHC class I molecules on their cells. Tatiana and Alexander expressed MHC class I molecules at very low levels, and when their isolated B cells were loaded with an antigenic peptide from influenza virus, this stimulated a small increase in the number of MHC class I molecules on these cells (Fig. 12.8). It is therefore possible that other virus infections that they sustained, such as chickenpox, were able to induce sufficient expression of MHC class I molecules to terminate the infection properly.

The repeated respiratory infections caused anatomic damage to their airways, resulting in the bronchiectasis. The abundant *Haemophilus* and pneumococci in their sputum is characteristic of patients with bronchiectasis, and in their case was not due directly to any deficiency of immunity against these capsulated bacteria (compare with Case 1). Patients with MHC class I deficiency often suffer from mid-line granulomatous disease, resembling Wegener's granulomatosis. Vasculitis is also common, especially on the extremities.

The profound reduction of CD8⁺ T cells in patients with MHC class I deficiency is a direct consequence of the lack of MHC class I molecules on the surface of epithelial and dendritic cells in the thymus, the organ in which all T cells bearing $\alpha\beta$ T-cell receptors mature. The interaction of thymocytes with MHC class I molecules expressed by thymic epithelial and dendritic cells is crucial for the intrathymic maturation of CD8⁺ T cells.

This family reveals that we have much to learn about the role of MHC class I molecules in protection against intracellular pathogens.

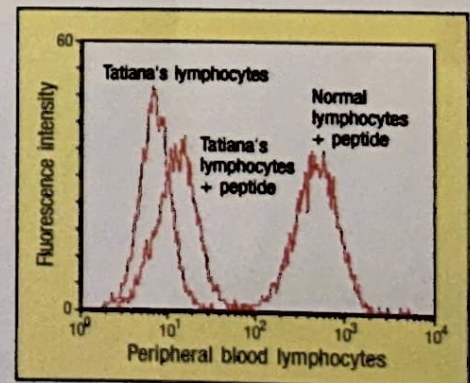


Fig. 12.8 Increased expression of MHC class I molecules in response to virus infection. Peripheral blood lymphocytes from Tatiana were examined with a fluorescent antibody against HLA-A3 before and after loading the cells with an antigenic peptide from influenza virus. There is a small increase in the number of MHC class I molecules expressed on the lymphocytes to approximately 5% of the amount expressed by normal cells loaded with the same peptide.

Questions.

- 1 Why was the number of $CD8^+$ T cells in Tatiana and Alexander decreased despite normal levels of $CD4^+$ T cells?
- 2 When the T-cell antigen receptors on their $CD8^+$ T cells were examined, it turned out that they were all $\gamma\delta$ receptors and none were $\alpha\beta$ receptors. How do you explain this?
- 3 Tatiana and Alexander had normal delayed-type hypersensitivity responses to tuberculin and Candida. Is this surprising in view of their deficiency in $CD8^+$ T cells?
- 4 Why did Tatiana and Alexander have high levels of serum IgG?
- 5 Genetic defects have also been found in the genes encoding TAP1 and TAPBP. Do you think that the clinical course in these patients would differ from that observed in Tatiana and Alexander?

CASE 13

X-LINKED LYMPHOPROLIFERATIVE SYNDROME

A defect in the immune response to a virus.

Viruses pose a constant challenge to our immune system. Unable to reproduce on their own, viruses have evolved as parasites, capable of residing within living cells whose biosynthetic machinery they subvert for their own reproduction. The human immune system combats these hidden, invading microbes through a two-pronged strategy: 1) elimination of viruses before they are able to enter host cells and 2) identification and destruction of infected host cells that harbor viruses. The innate and adaptive immune systems are both enlisted in the defense against viruses.

Innate pattern recognition receptors, including the endosomal toll-like receptors (TLRs) 3, 7, 8, and 9 along with cytosolic RIG-like receptors (RLRs), sense intracellular viral particles by identifying prokaryotic nucleic acid fragments within the host cell. Activated TLRs and RLRs induce signaling through the interferon regulatory factor (IRF) family of transcription factors to produce type I interferons (most importantly, IFN α and IFN β). These interferons engender an antiviral state within the host by promoting transcription of genes that resist viral infection, increasing lymphocyte retention in the lymph nodes, favoring T_H1 differentiation, and enhancing the cytotoxic effector functions of natural killer (NK) and CD8 T cells.

NK cells of the innate immune system also combat viral infections through the cytotoxic response, eliminating infected host cells (Fig. 13.1). NK cells are large granular cells of the lymphocyte lineage; however, they differ from the adaptive immune system's B and T cell lymphocytes in that they express only germline DNA receptors. Hence, these cells are equipped to react immediately upon the first encounter with the virus and do not require activation, clonal expansion, and differentiation into effector cells like the cytotoxic T lymphocytes (CTLs). NK cells constantly survey host cells with telltale markers of viral infection. NK cells express receptors that activate or inhibit the cell's killer activity. The fate of potential target cells is determined by the balance of activating and inhibitory signals delivered

TOPICS BEARING ON THIS CASE:

NK-cell activating receptors and killer function

NK T cells

Cell signaling from tyrosine kinase-associated receptors

Activation and function of cytotoxic T cells